

DAC WAVEFORM GENERATION PROCEDURE

Note: Remove R919, Probe U28.20 and check for 2.5V.

1. Write Data = 0x00000004 in address = 0x10001004.
2. Probe and check U30.1 = GND, U30.14 = 3.3V & U30.2 = 3.3V.
→ If not found OK, Don't proceed further.
3. Write Data = 0x00000001 in address = 0x10001000
4. Wait for 2ms, for configuration to be completed.
5. Write Data = 0x00007FFF in address = 0x10001008.
6. Write Data = 0x00553261 in address = 0x1000100C
7. Write Data = 0x00000055 in address = 0x10001010.
8. Observe sine wave on pin U29.6.

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DAC TESTING.

- Hardware Modification →
1. Remove R919, R346 for channel 1. [Apply same for 2 to 12]
 2. Populate R345 (10k) for channel 1. [——— " ———]

Software Procedure

1. Set output mux for voltage range -10V to +10V. So write data = 0x04 in address = 0x10001004
2. Start DAC fixed configuration. Write data = 0x01 in add = 0x10001000.
3. Wait for 120ms.

Static DC voltage :
(a) Write output voltage DAC converted value (16 bit) in address = 0x10001008
(b) Write DC voltage gen. command 0x56 in address = 0x10001010

Sine Wave generation:

Modulation index formula

$$M = \frac{f_{out} \times 2^{32}}{f_{REF}}$$

$f_{REF} = 1\text{MHz}$ & f_{out} = Required frequency.

- (a) Write amplitude 0x7FFF in address = 0x10001008.
- (b) Write mod. index 0x00553261 in address = 0x1000100C.
- (c) Write sine gen command 0x55 in address = 0x10001010.